

Vector algebra

2.2.5 GRADIENT OF A SCALAR FIELD

The *gradient* of p , ∇p , at a given point in space is defined as a vector such that:

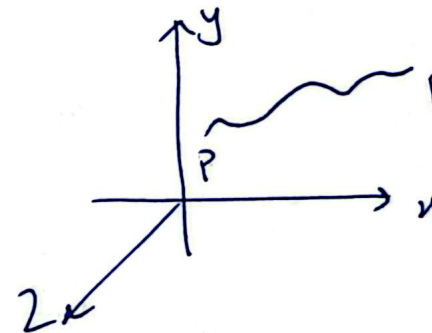
1. Its magnitude is the maximum rate of change of p per unit length of the coordinate space at the given point.
2. Its direction is that of the maximum rate of change of p at the given point.

scalar $\leftarrow p = p(x, y, z)$

Gradient of p (scalar) is a vector

$$\nabla p = \frac{\partial p}{\partial x} \mathbf{i} + \frac{\partial p}{\partial y} \mathbf{j} + \frac{\partial p}{\partial z} \mathbf{k}$$

$p = p(r, \theta, z)$



Gradient of p in cylindrical coordinate

$$\nabla p = \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{\partial p}{\partial z} \mathbf{e}_z$$

$p = p(r, \theta, \Phi) \rightarrow$ "Spherical coordinate"

$$\nabla p = \frac{\partial p}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial p}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial p}{\partial \Phi} \mathbf{e}_\Phi$$

Vector algebra

2.2.6 DIVERGENCE OF A VECTOR FIELD

Read the next

Cartesian:

$$\mathbf{V} = \mathbf{V}(x, y, z) = V_x \mathbf{i} + V_y \mathbf{j} + V_z \mathbf{k}$$

Divergence $\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$

Cylindrical:

$$\mathbf{V} = \mathbf{V}(r, \theta, z) = V_r \mathbf{e}_r + V_\theta \mathbf{e}_\theta + V_z \mathbf{e}_z$$

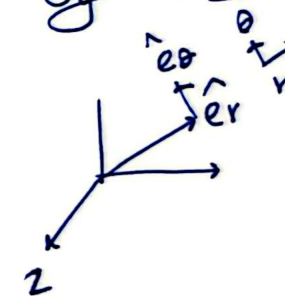
$$\nabla \cdot \mathbf{V} = \frac{1}{r} \frac{\partial}{\partial r} (r V_r) + \frac{1}{r} \frac{\partial V_\theta}{\partial \theta} + \frac{\partial V_z}{\partial z}$$

Spherical:

$$\mathbf{V} = \mathbf{V}(r, \theta, \Phi) = V_r \mathbf{e}_r + V_\theta \mathbf{e}_\theta + V_\Phi \mathbf{e}_\Phi$$

$$\nabla \cdot \mathbf{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (V_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial V_\Phi}{\partial \Phi}$$

Ex: $\vec{V} = 2r \hat{e}_r + r \sin \theta \hat{e}_\theta$
Is this velocity physical for incomp. flow?



A scalar quantity given as a function of coordinate space and time t is called a *scalar*
vector quantity given as a function of coordinate space and time is called a *vector*

Solution: To show it is physical, $\vec{\nabla} \cdot \vec{v} = 0$

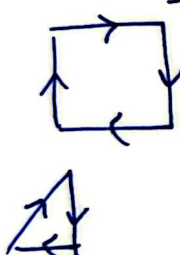
$$\vec{\nabla} \cdot \vec{v} = \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} = \frac{1}{r} \frac{\partial}{\partial r} (r \times \underbrace{2r}_{2r^2}) + \frac{1}{r} \frac{\partial (r \sin \theta)}{\partial \theta} = 4 + \cos \theta \neq 0$$

Not physical!

Relations Between Line, Surface, and Volume Integrals

$$\Gamma = \oint_C \mathbf{V} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S}$$

length
area



Consider again the open area S bounded by the closed curve C , as shown in Figure 2.11. Let $\mathbf{A}$ be a vector field. The line integral of $\mathbf{A}$ over C is related to the surface integral of $\mathbf{A}$ over S by *Stokes' theorem*:

$$\oint_C \mathbf{A} \cdot d\mathbf{s} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S} \quad (2.25)$$

Consider again the volume V enclosed by the closed surface S , as shown in Figure 2.12. The surface and volume integrals of the vector field $\mathbf{A}$ are related through the *divergence theorem*:

$$\oiint_S \mathbf{A} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{A}) dV \quad (2.26)$$

If p represents a scalar field, a vector relationship analogous to Equation (2.26) is given by the *gradient theorem*:

$$\oiint_S p d\mathbf{S} = \iiint_V \nabla p dV \quad (2.27)$$



PHYSICAL MEANING OF THE DIVERGENCE OF VELOCITY

divergence theorem, Equation (2.26), to the right side of Equation (2.29), we have

$$\frac{D\mathcal{V}}{Dt} = \iiint_{\mathcal{V}} (\nabla \cdot \mathbf{V}) d\mathcal{V} \quad [2.30]$$

Assume that $\delta\mathcal{V}$ is small enough such that $\nabla \cdot \mathbf{V}$ is essentially the same value throughout $\delta\mathcal{V}$. Then the integral in Equation (2.31) can be approximated as $(\nabla \cdot \mathbf{V})\delta\mathcal{V}$. From

$$\frac{D(\delta\mathcal{V})}{Dt} = (\nabla \cdot \mathbf{V})\delta\mathcal{V}$$

$$\boxed{\nabla \cdot \mathbf{V} = \frac{1}{\delta\mathcal{V}} \frac{D(\delta\mathcal{V})}{Dt}}$$

Examine Equation (2.32). It states that $\nabla \cdot \mathbf{V}$ is physically the time rate of change of the volume of a moving fluid element, per unit volume. Hence, the interpretation of $\nabla \cdot \mathbf{V}$, first given in Section 2.2.6, Divergence of a Vector Field, is now proved.

$m\vec{V}$: Momentum

Force $\equiv$ rate of change of momentum

~~transport of~~
~~Conservation of Momentum~~ (TSec. 2.3)

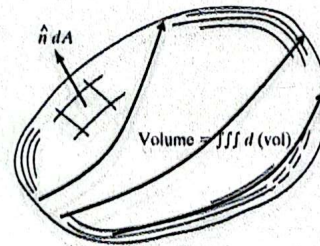


Figure 1.6: modified Fig. 2.5 from AE-BC

Force results in momentum change.

$$\left\{ \begin{array}{l} \text{If } \vec{F} = 0 \text{ then} \\ 0 = \frac{d}{dt} (m\vec{V}) \end{array} \right.$$

State of conservation of momentum: Newton's second law.
Mathematically expressed as

$$\vec{F} = m\vec{a} = \frac{d}{dt} (m\vec{V}) \quad (1.14)$$

where $\vec{F}$ is the force on the fluid, $\vec{a}$ the acceleration and m the mass.

transport of Conservation of Momentum

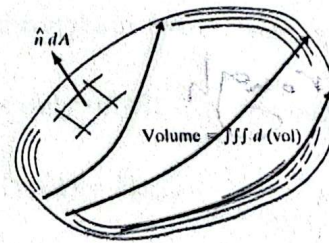


Figure 1.7: modified Fig. 2.5 from AE-BC

$$\begin{aligned}
 & \{\text{rate of momentum accumulation in (vol)}\} \\
 &= \{\text{rate of momentum in}\} - \{\text{rate of momentum out}\} + \\
 & \quad \{\text{sum of forces acting on the system}\} \\
 &= -\{\text{net rate of momentum flow out of (vol)}\} \\
 & \quad + \{\text{sum of forces acting on the system}\} \quad (1.15)
 \end{aligned}$$

Cons. of Mass

rate of change of
mass in volume (Vol.)

+

mass transport through
the surfaces

||

0

Transport of Mom.

rate of change
of Mom ($m\bar{v}$) in Vol.

+

Mom. transport through
the surfaces

||

Force

{	body force
	surface force

Force → ① Body force, assume body force per ~~volume~~^{mass} is $\vec{F}_b$
 So total body force is $\iiint (\rho \vec{F}_b d\text{vol})$
 → ② Surface force:
 Example of body force is "weight"^{dm}
 like pressure, viscous force, i.e., pressure

Applied forces

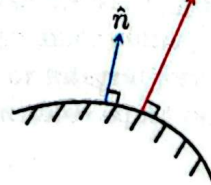
$$\iint_A p \hat{n} \cdot d\vec{A}$$

- ▶ Body forces - acts throughout (vol), e.g., gravity, electromagnetic forces
- ▶ Surface forces - acts only on A, e.g., pressure, viscous forces

$$\{\text{body force on (vol)}\} = \vec{F}_{\text{body}} = \iiint_{(\text{vol})} \rho \vec{F}_b d(\text{vol}) \quad (1.16)$$

where, ρ is mass/unit volume and $\vec{F}_b$ is force/unit area.

$$\{\text{surface force on (vol)}\} = \vec{F}_{\text{surface}} = \iint_A f_s \hat{n} \cdot d\vec{A} \quad (1.17)$$



Continuity: $\frac{\partial}{\partial t} \iiint \rho \, d(\text{Vol}) + \iint \rho \vec{V} \cdot \vec{n} \, dA = 0$

Momentum tra.: $\underbrace{\frac{\partial}{\partial t} \iiint \rho \vec{V} \, d(\text{Vol})}_{\text{Momentum change in C.V.}} + \underbrace{\iint \rho \vec{V} \vec{V} \cdot \vec{n} \, dA}_{\text{momentum transport through the surfaces}} = - \iint p \hat{n} \cdot dA + \iiint \rho \vec{F}_b \, d(\text{Vol})$
 $+ F_{\text{viscous}} + F_{\text{ext}}$

Conservation of momentum

Using eq. (1.16) and (1.17) in eq. (1.15) we get,

$$\underbrace{\frac{\partial}{\partial t} \iiint_{\text{vol}} \rho \vec{V} \, d(\text{vol})}_{\text{accumulation}} = - \underbrace{\iint_A \rho \vec{V} \, dA}_{\text{I}} \underbrace{(\vec{V} \cdot \hat{n})}_{\text{II}} - \iint_A p \hat{n} \cdot dA + \vec{F}_{\text{visc}} + \vec{F}_{\text{ext}} + \iiint_{\text{vol}} \rho \vec{f}_b \, d(\text{vol}) \quad (1.18)$$

where, I - quantity being transported, momentum ($\rho \vec{V}$) and II - velocity at which quantity is being transported. Here the momentum is being transported at a velocity $\vec{V} \cdot \hat{n}$. Also, p is the pressure, $\vec{F}_{\text{visc}}$ is the viscous forces, $\vec{F}_{\text{ext}}$ is any other external force that may be acting, and $\vec{f}_b$ (per unit volume) is any body force that may be acting. This is the **global** or **integral** form of the momentum equations. It's a vector equation with three equations when considering 3-D flow.

MOMENTUM EQUATION

$$\frac{\partial}{\partial t} \iiint_V \rho \mathbf{V} dV + \iint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V} = - \iint_S p d\mathbf{S} + \iiint_V \rho \mathbf{f} dV + \mathbf{F}_{\text{viscous}}$$

Momentum Eq.

$$- \iint_S p d\mathbf{S} = - \iiint_V \nabla p dV \quad \text{Gradient Theorem} \quad (1)$$

$$\iiint_V \frac{\partial(\rho \mathbf{V})}{\partial t} dV + \iint_S \underbrace{(\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V}}_{\text{mass} \times \mathbf{V}} = - \iiint_V \nabla p dV + \iiint_V \rho \mathbf{f} dV + \mathbf{F}_{\text{viscous}}$$

[2.66]

the x component of Equation (2.66) is

Do not decompose mass!

x-Mom

$$\iiint_V \frac{\partial(\rho u)}{\partial t} dV + \iint_S \underbrace{(\rho \mathbf{V} \cdot d\mathbf{S}) u}_{\text{mass} \times u} = - \iiint_V \frac{\partial p}{\partial x} dV + \iiint_V \rho f_x dV + (F_x)_{\text{viscous}}$$

divergence theory [2.67]

$$\iint_S (\rho \mathbf{V} \cdot d\mathbf{S}) u = \iint_S (\rho u \mathbf{V}) \cdot d\mathbf{S} = \iiint_V \nabla \cdot (\rho u \mathbf{V}) dV$$

(2)

theory

x-Mom

$$\iiint_V \left[\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) + \frac{\partial p}{\partial x} - \rho f_x - (F_x)_{\text{viscous}} \right] dV = 0$$

MOMENTUM EQUATION

"Differential format"

x-Mom

$$\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \mathbf{V}) = -\frac{\partial p}{\partial x} + \rho f_x + (\mathcal{F}_x)_{\text{viscous}}$$

y-Mom

$$\frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v \mathbf{V}) = -\frac{\partial p}{\partial y} + \rho f_y + (\mathcal{F}_y)_{\text{viscous}}$$

z-Mom

$$\frac{\partial(\rho w)}{\partial t} + \nabla \cdot (\rho w \mathbf{V}) = -\frac{\partial p}{\partial z} + \rho f_z + (\mathcal{F}_z)_{\text{viscous}}$$

Note that Equations (2.64) and (2.70a to c) apply to the unsteady, three-dimensional flow of any fluid, compressible or incompressible, viscous or inviscid. Specialized to a steady ($\partial/\partial t \equiv 0$), inviscid ($\mathcal{F}_{\text{viscous}} = 0$) flow with no body forces ($\mathbf{f} = 0$), these equations become

If the flow is
steady, $\frac{\partial}{\partial t} = 0$
and

$$\oint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V} = - \oint_S p d\mathbf{S}$$

[2.71]

If the flow is inviscid
w.o. body force:

$$f_x = f_y = f_z = 0$$

$$(\mathcal{F}_x)_v = (\mathcal{F}_y)_v = (\mathcal{F}_z)_v = 0$$

$$\nabla \cdot (\rho u \mathbf{V}) = -\frac{\partial p}{\partial x}$$

[2.72a]

$$\nabla \cdot (\rho v \mathbf{V}) = -\frac{\partial p}{\partial y}$$

[2.72b]

$$\nabla \cdot (\rho w \mathbf{V}) = -\frac{\partial p}{\partial z}$$

[2.72c]

steady, no body force, no viscous force:

$$x\text{-Mom: } \nabla \cdot (\rho u \vec{V}) = -\frac{\partial p}{\partial x}$$

$$\nabla \cdot () = \frac{\partial}{\partial x} ()_x + \frac{\partial}{\partial y} ()_y + \frac{\partial}{\partial z} ()_z$$

$$x\text{-Mom} \quad \nabla \cdot (\rho u \vec{V}) = \frac{\partial}{\partial x} (\rho u u) + \frac{\partial}{\partial y} (\rho u v) + \frac{\partial}{\partial z} (\rho u w) = -\frac{\partial p}{\partial x}$$

Assume flow is 1-D, $\{v=w=0\}$ the above Eqⁿ simplifies

$$\text{to: } \frac{\partial}{\partial x} (\rho u u) = -\frac{\partial p}{\partial x} \quad \text{or} \quad \frac{\partial}{\partial x} (\rho u^2 + p) = 0$$

$$\rho u^2 + p = \text{cte} \quad \text{or} \quad \rho_1 u_1^2 + p_1 = \rho_2 u_2^2 + p_2$$

Integrated form of momentum Eq.

Note for incomp flow, we had Bernoulli Eqⁿ Ber.

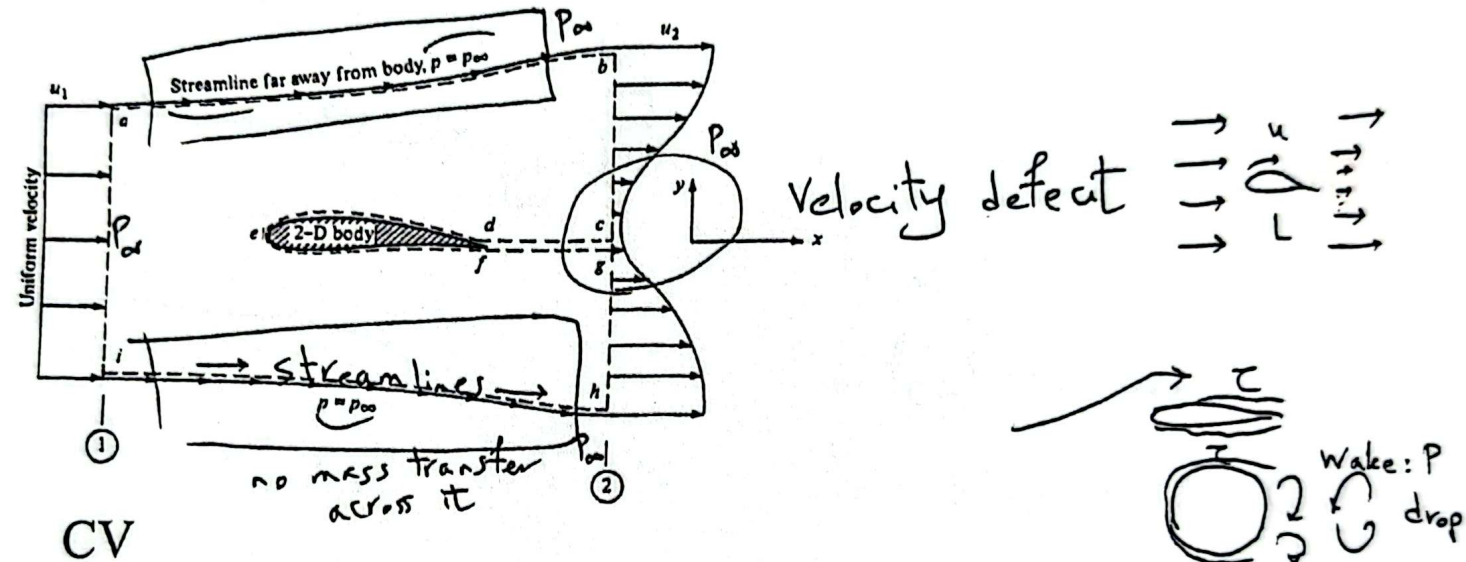
$p_1 + \rho_1 u_1^2 = p_2 + \rho_2 u_2^2$ Comp.

$\frac{1}{2} \rho u_1^2 + p_1 = \frac{1}{2} \rho u_2^2 + p_2$

for comp. flow, we have:

The C.V excludes the airfoil as it is solid object

AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

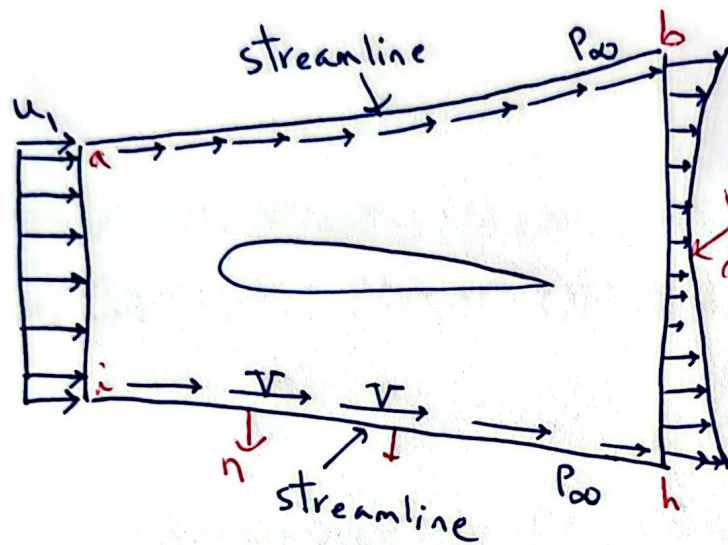


1. The upper and lower streamlines far above and below the body (ab and hi, respectively).
2. Lines perpendicular to the flow velocity far ahead of and behind the body (ai and bh, respectively).
3. A cut that surrounds and wraps the surface of the body (cdefg).

$$\vec{V} \cdot \vec{n} \big|_{outlet}$$

$$\vec{V} \cdot \vec{n} \big|_{inlet}$$

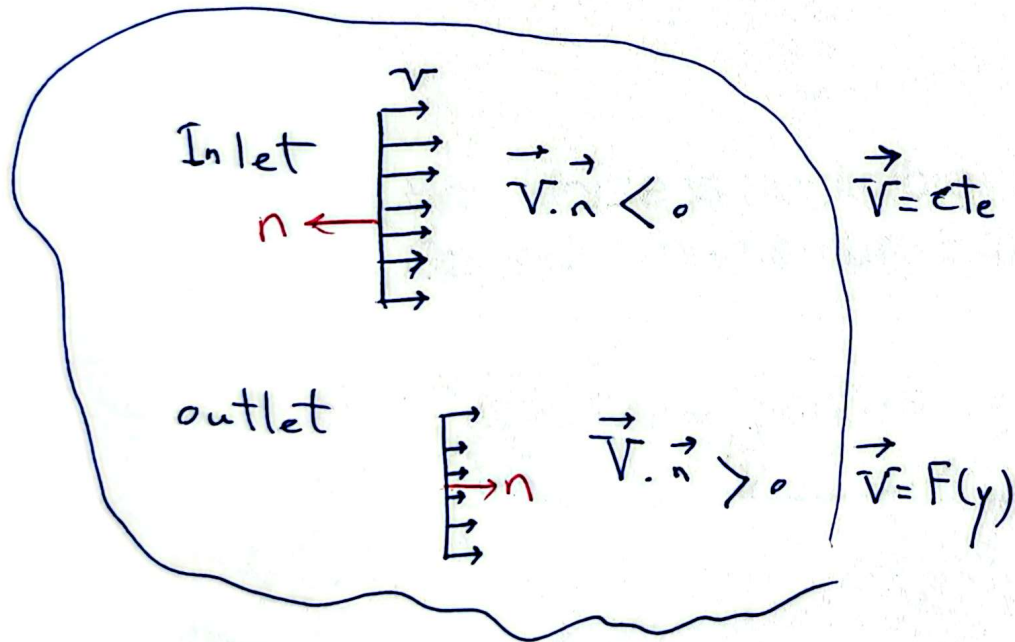
$$\iint_S \vec{V} \cdot \vec{n} : mass$$



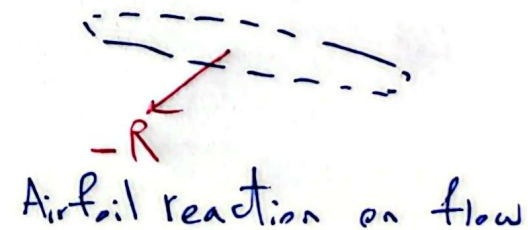
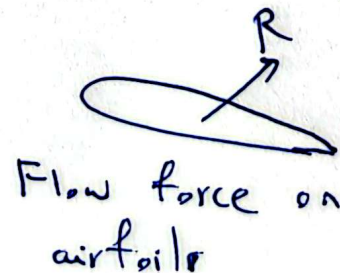
streamline is a line tangent to velocity field!

velocity deficit $\vec{V} \cdot \vec{n} = 0 \rightarrow$ no mass transfer across the streamline

over the airfoil, boundary layer forms and we have velocity deficit in some regions, so to keep mass conserved, area increases (streamlines diverge)



We can remove airfoil and put its resultant force instead



AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

$$\text{Surface force} = - \underbrace{\iint_{abhi} p \, dS - \underline{\underline{R'}}}_{\text{Forces}} \quad [2.73]$$

Body force is negligible

Consider momentum Equation in integral form

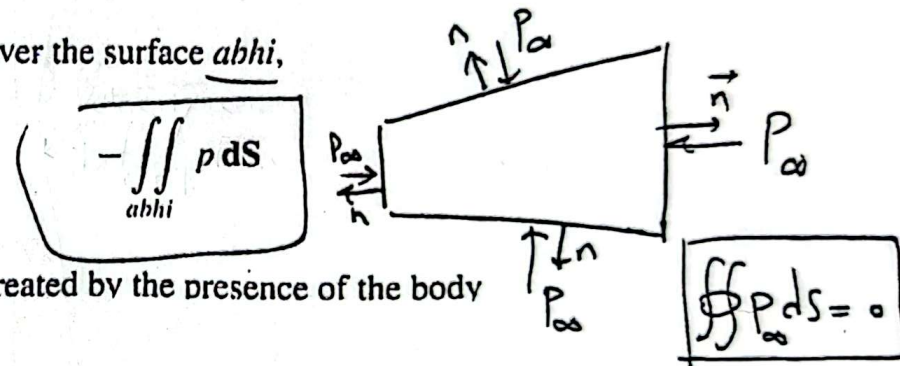
$$\underbrace{\cancel{\frac{\partial}{\partial t} \iiint_V \rho \mathbf{V} \, dV}}_{\text{steady flow}} + \underbrace{\iiint_S (\rho \mathbf{V} \cdot d\mathbf{S}) \mathbf{V}}_{\text{Forces}} = - \underbrace{\iint_{abhi} p \, dS - \underline{\underline{R'}}}_{\text{Forces}} \quad [2.74]$$


AN APPLICATION OF THE MOMENTUM EQUATION: DRAG OF A TWO-DIMENSIONAL BODY

$P = P_{\infty}$ on the CV $U_1 = Cte., U_2 = F(y)$

Consider the surface forces on the control volume shown in Figure 2.18a. They stem from two contributions:

1. The pressure distribution over the surface $abhi$,



2. The surface force on def created by the presence of the body

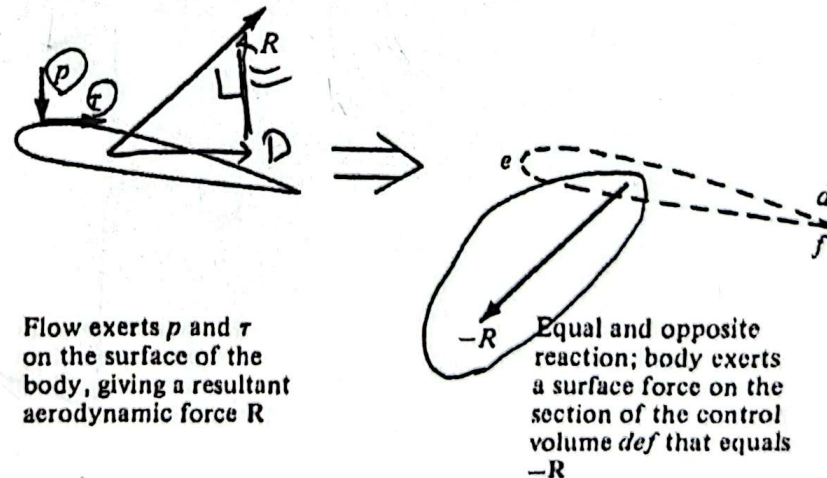


Figure 2.19 Equal and opposite reactions on a body and adjacent section of control surface.

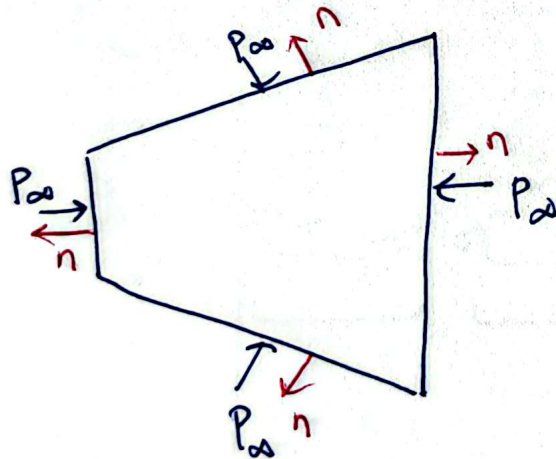
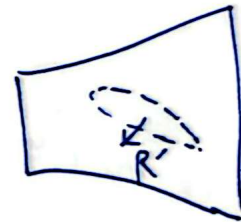
Forces are pressure force and $-R'$

$$\Rightarrow F = - \int_{\text{abhi}} p ds - R' \quad \underline{\text{ignore body force}}$$

write Mom. Eqⁿ in integral format

$$\cancel{\frac{\partial}{\partial t} \iiint \rho \mathbf{V} d\mathbf{v}} + \int_S (\rho \mathbf{V} \cdot d\mathbf{s}) \mathbf{V} = - \underbrace{\int_S p d\mathbf{s} - R'}_{\text{force}}$$

steady flow



pressure